

Christmas Party

Time limit: 1.00 s

Memory limit: 512 MB

There are n children at a Christmas party, and each of them has brought a gift. The idea is that everybody will get a gift brought by someone else.

In how many ways can the gifts be distributed?

Input

The only input line has an integer n : the number of children.

Output

Print the number of ways modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 10^6$

Example

Input	Output
4	9